



## PROJECT OVERVIEW

# Project Overview

Architecture, controls, security, deployment, and project status.

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## Move your hand. Play the game.

VirtualGlove turns hand movement and gestures into responsive RetroArch controls using an ordinary USB camera and a **VirtualGlove Controller** built on the Arduino UNO Q. Wear a plain glove or use your bare hand-there are no sensors, wires, or electronics to add to it.

Move to steer. Curl fingers for buttons. Roll, push, pull, grab, throw, and punch. VirtualGlove recognizes the pose, sends authenticated controller input across your local network, and lets RetroArch see controller input or a native Power Glove controller.

**Current project version: 0.5.0 · Next candidate: v0.5.0-rc.1**

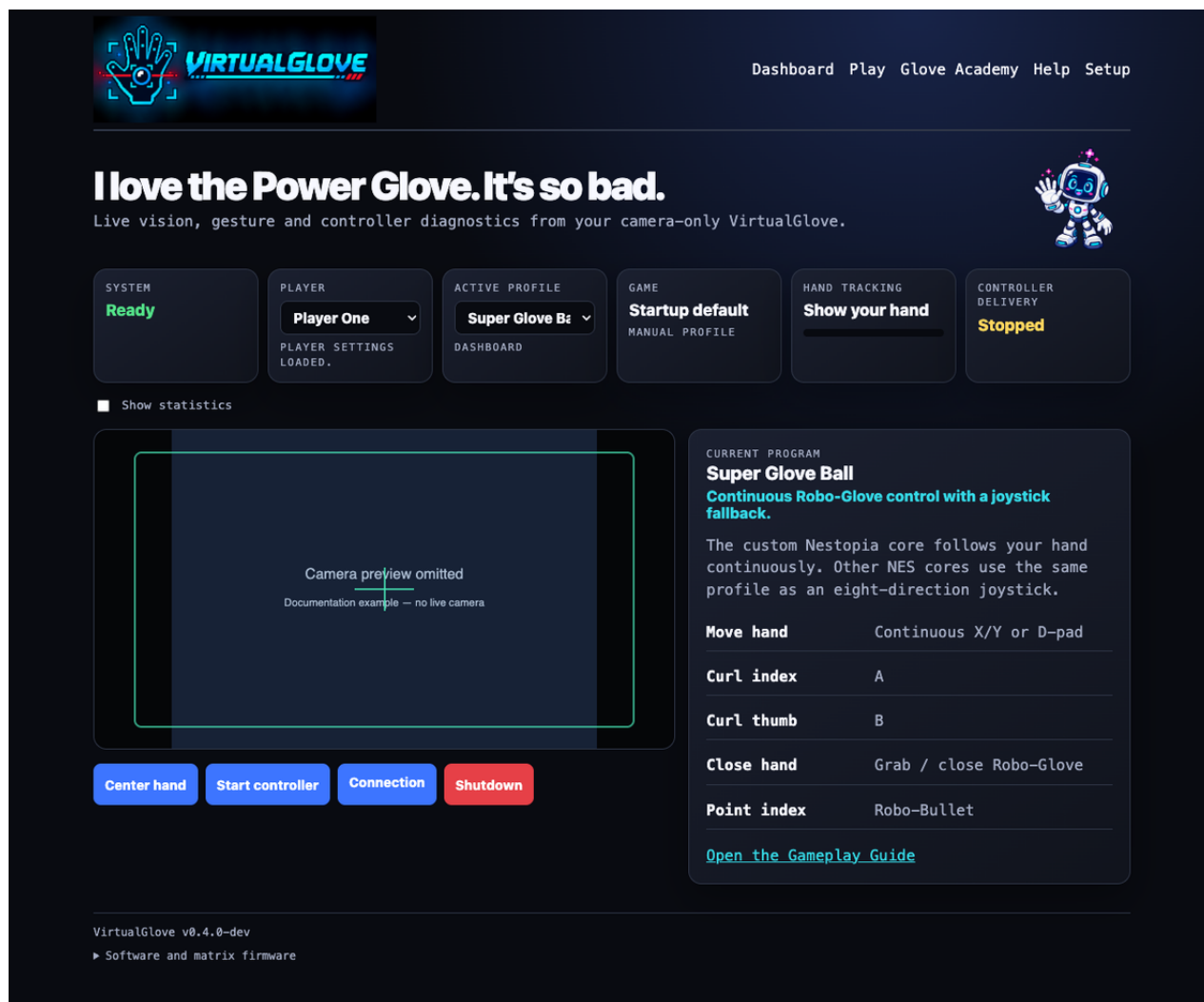
VirtualGlove supports RetroPie, Recalbox 10.x, Batocera 38+, and LaunchBox on 64-bit Windows. Recalbox and Batocera use **Controller Router** to combine configured physical controllers and VirtualGlove as enabled merged Players 1-4. RetroPie can opt into the same Router while retaining its separate-gamepad default. LaunchBox keeps physical XInput beside VirtualGlove's managed RetroArch controller. Every platform keeps the physical controller usable and preserves unrelated controller settings.

When Controller Router is enabled, a game launch always begins neutral. Physical controllers are available immediately; VirtualGlove joins only after the hand has returned to neutral once. This prevents a gesture observed in EmulationStation from becoming the game's first input.

Ordinary NES games use FCEUmm. Super Glove Ball can also use the separately named Nestopia (VirtualGlove) core for native movement and glove actions without replacing stock Nestopia. The v0.5.0 installers provide a guarded upgrade from v0.4.2 while preserving players, calibration, tuning, pairing, device settings, game registrations, ROMs, saves, and controller assignments.

## Why VirtualGlove?

- **Camera-only play:** use a standard UVC camera rather than modifying a glove.
- **Two styles of NES control:** ordinary joystick output through FCEUmm and continuous native movement for Super Glove Ball through [lr-nestopia-powerglove](#).
- **Original programs and game mappings:** Programs 1-14 and A-I, including Mattel's indexed games and documented rapid-fire exceptions.
- **Fast, direct tracking:** MediaPipe Hands processes the newest camera frame and sends the latest valid hand coordinate without a settling tail.
- **Family-friendly learning:** Pixel Pal guides players through 16 Glove Academy lessons without sending accidental input to a game.
- **Personal setup:** each player can save a hand centre, movement reach, joystick centre box, gesture sensitivity, and Academy progress.
- **Game-aware and safe:** registered games select their profile automatically; stale, lost, or unauthenticated input returns to neutral.
- **Local by design:** video stays on the Controller. Pairing and controller traffic are authenticated between your own devices.



VirtualGlove Dashboard with camera and controller status

## What you need

- An Arduino UNO Q provisioned through Arduino App Lab
- A supported console: RetroPie, Recalbox 10.x, Batocera 38+, or LaunchBox with 64-bit Windows and 64-bit RetroArch
- A UVC-compatible USB camera and powered USB hub
- A physical controller for RetroArch setup and recovery
- Both devices on the same trusted local network with internet access during installation
- Your own legally obtained games-VirtualGlove includes no ROMs or BIOS files

New to the hardware? Start with [Build your own](#) for the parts, expected cost, and difficulty.

# Install VirtualGlove

These steps install the latest stable release. Install the **same version on both devices** and close any running RetroArch game first. The scripts verify their downloads, ask for administrator access when needed, and preserve existing pairing and player settings during an update.

Release-candidate testers should install **v0.5.0-rc.1** on both machines using the pinned-version procedure in the [technical installation reference](#). The unversioned commands below deliberately continue to select the latest stable release.

## 1. Prepare the Controller

Finish the UNO Q's App Lab setup, connect it to your network, and attach the camera through the powered hub. The camera may also be connected after installation.

## 2. Install the Controller software

Open a terminal on the UNO Q and run:

```
SH
cd /home/arduino
curl -fLO \
  https://github.com/mathan416/VirtualGlove/releases/latest/download/install-uno-q.sh \
  && bash install-uno-q.sh
```

On a first installation, the installer suggests **virtualglove** as the Controller name, making its usual address **virtualglove.local**. Press Enter to accept it or type a different family-friendly name. Updates preserve the existing name.

## 3. Check the Controller

Open the Dashboard address printed by the installer—normally <http://virtualglove.local:8088/dashboard>. The installer configures automatic startup, the matrix display, guarded camera recovery, and its required host helpers.

## 4. Install the console software

Choose exactly one console installer:

| Console   | Run on the console                                           | Installed in                                                                |
|-----------|--------------------------------------------------------------|-----------------------------------------------------------------------------|
| RetroPie  | <a href="#">install-retro pie.sh</a> as the normal user      | <a href="#">/opt/virtualglove-src</a> and <a href="#">/etc/virtualglove</a> |
| Recalbox  | <a href="#">install-recal box.sh</a> as root                 | <a href="#">/recalbox/share/system/virtualglove</a>                         |
| Batocera  | <a href="#">install-batoc era.sh</a> as root                 | <a href="#">/userdata/system/virtualglove</a>                               |
| LaunchBox | <a href="#">install-launc hbox.ps1</a> as the Windows player | <a href="#">%LOCALAPPDATA%\VirtualGlove</a>                                 |

Each Linux command downloads the latest installer from GitHub and selects the latest stable release automatically. LaunchBox is installed from the extracted Windows package. The complete [Installation Guide](#) has copyable commands, prerequisites, questions, and checkpoints for all four platforms.

Each installer uses the platform's persistent storage, preserves ROMs, saves, the game registry, pairing, and unrelated controller configuration, and keeps a physical Player 1 joypad usable beside VirtualGlove. Batocera automatically selects the native core only for exact registered Super Glove Ball filenames that do not already have an explicit core choice. A target without a loadable native core still receives the complete FCEUmm joystick path.

The generic RetroPie installation exposes a separate **VirtualGlove** gamepad. The project-maintained [arcade-cabinet-merger](#) remains as the development cabinet's proven reference and rollback implementation. The cabinet now runs the shared Controller Router, which generalizes that work without changing RetroPie's default. Recalbox and Batocera create the enabled **VirtualGlove Merged Player 1-4** devices and initially preserve the installer-selected Player 1 arrangement. Setup can then assign several configured physical sources to a player and place the single VirtualGlove on exactly one player. Original controllers remain the only active frontend controllers. LaunchBox keeps physical XInput plus a loopback Network RetroPad for FCEUmm games; its real keyboard mappings remain available as a manual fallback. Native Super Glove Ball uses only the guarded Power Glove state channel; it does not duplicate native gestures as keyboard events. Its installer makes **VirtualGlove RetroArch** the default NES emulator after backing up LaunchBox's emulator and NES game definitions. Existing NES games assigned to standard RetroArch and new NES imports use VirtualGlove automatically; games assigned to a genuinely different emulator remain explicit overrides.

## 5. Pair the devices

Open the secure Setup address printed by the installer—normally <https://virtualglove.local:8443/setup>—open **Connection and startup**, choose RetroPie, Recalbox, Batocera, or LaunchBox, save the console hostname or IP address, and continue into **Pair this Controller**. Pairing stays unavailable until the platform and address are both saved. The guided one-time-code method is recommended and shows only the command for the selected platform. For an optional, resumable first-game walkthrough, select **Get ready to play** on Setup or Dashboard. It checks your player, connection, camera, center, and essential gestures in safe practice before you explicitly enable game controls.

After confirming the browser certificate against the physical Matrix ID, the optional **Trust this Controller** step removes future privacy warnings on that phone or computer.

## 6. Set up a player

Choose a player, position the camera, and use **Center hand**. Open **Glove Academy** to learn the gestures and adjust movement reach or sensitivity only if needed.

## 7. Play

Select **Start controller**, launch a registered game, and confirm the expected profile. FCEUmm uses joystick mode; Super Glove Ball can also use the optional native core selected per ROM on RetroPie, Recalbox, Batocera, or LaunchBox.

ROMs added later do not require reinstalling VirtualGlove. Refresh the console frontend's game list, then open **Setup -> Games** and register the exact ROM filename if it is not already one of the supplied aliases. RetroPie uses its per-ROM launch choice for native Super Glove Ball; Recalbox and Batocera apply their exact-ROM native choice at VirtualGlove service startup; LaunchBox's VirtualGlove RetroArch wrapper chooses the core on every launch. The complete installation guide lists the platform-specific refresh and restart steps.

The complete [Installation Guide](#) has first-install checkpoints, illustrated pairing, camera advice, native-core setup, updates, backups, and troubleshooting. Use it as the authoritative setup reference.

## What can you play?

VirtualGlove includes the original Programs 1-14, nine reusable cartridge Programs A-I, plus dedicated mappings for Bad Street Brawler and Super Glove Ball. The same recognition settings follow the player across games; profiles change only what the recognized movements and gestures send to the console.

| Path                                          | What it provides                                                                                                                                                                                              |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FCEUmm                                        | The selected Programs 1-14 or A-I mapping, including positional movement, mapped A/B actions, and documented compound gestures.                                                                               |
| Super Glove Ball with FCEUmm                  | A complete joystick-mode fallback that can always be selected for testing or play.                                                                                                                            |
| Super Glove Ball with Nestopia (VirtualGlove) | Continuous native X/Y and Z, Start, grab/catch, release/throw, Robo-Bullet fire, and Power Punch. The isolated core is supported on RetroPie, Recalbox, Batocera, and LaunchBox; FCEUmm remains the fallback. |

Rapid A/B defaults on only where an individual program description explicitly identifies a pulsed button: Program 7 A, Program B A, Program H A/B, and Bad Street Brawler B. Every other profile defaults off. The registered Blaster Master, Double Dribble, Racket Attack, Ice Hockey, and Alpha Mission entries retain Mattel's documented off instructions. Programs 13 and 14 also support deliberate physical-controller and gestures-off play.

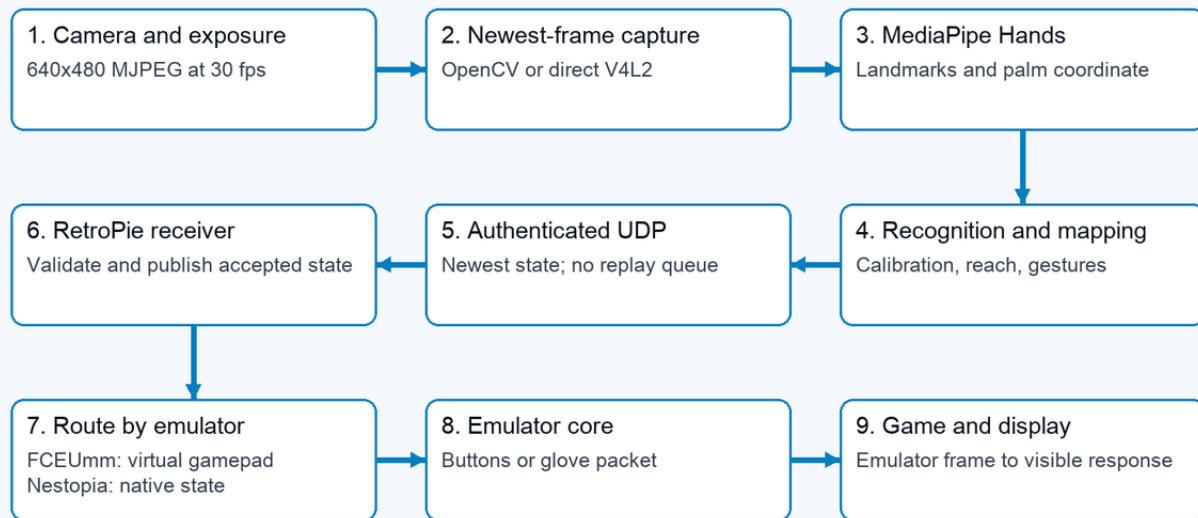
Dashboard Rapid A/B switches control button repetition only; profile-owned fast turns, pulsed movement, turbo movement, and compound actions retain their own timing. Saved per-game overrides survive upgrades. Choose **Use profile defaults** while that game is running to remove its overrides and adopt the corrected defaults.

The [Gameplay Guide](#) shows every numeric Program gesture, compound action, release rule, indexed game, exception, objective, and practice challenge. The [VirtualGlove Input Modes](#) explains why the joystick and native emulator paths feel different and how physical Player 1 controls join each platform.

The numeric Program section includes gesture-by-gesture controls, compound action timing, the complete official game index, and the five automatic rapid-fire exceptions. Program 13 is the mixed physical-controller option; Program 14 deliberately turns camera gestures off for the active game.

# How it works

## 08 / Camera to game: one front end, two delivery paths



The browser preview and statistics observe the worker; neither belongs in the controller delivery path.

*End-to-end VirtualGlove flow from camera to game*

1. The camera delivers its newest frame to the VirtualGlove Controller.
2. MediaPipe Hands finds the palm, wrist, and finger landmarks.
3. Shared recognition turns those landmarks into position, fingers, rolls, depth motion, and menu poses.
4. The Controller sends the newest authenticated state to the selected console.
5. The console publishes either Player 1 input or native glove state for the selected emulator core.

Pairing is tied to a private shared key rather than one permanent IP address. If DHCP changes an address or `.local` resolution temporarily fails, the paired devices can rediscover one another on the local network without broadcasting controller states or requiring a new pairing.

## Meet Pixel Pal

Pixel Pal helps players learn, personalize, test, and troubleshoot without turning setup into an engineering exercise.

- **Glove Academy** teaches all 16 movements and gestures while game output is paused.
- **Tune gestures** records guided examples and previews a conservative adjustment before saving it.
- **Find the best camera settings** compares only choices supported by the attached camera and changes nothing until the player accepts a recommendation.
- **Rock Paper Scissors** provides a camera-controlled practice game that does not require a connected console.

## Controller pages

| Page                                   | Purpose                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------|
| Dashboard · <a href="#">/dashboard</a> | See live controls and set per-game A/B rapid fire.                                        |
| Play · <a href="#">/play</a>           | Challenge Pixel Pal to Rock Paper Scissors.                                               |
| Glove Academy · <a href="#">/learn</a> | Learn gestures, set movement reach, and personalize recognition safely.                   |
| Setup · <a href="#">/setup</a>         | Manage players, camera choices, console pairing, games, backups, and display preferences. |
| Help · <a href="#">/help</a>           | Read the complete manuals and printable PDFs directly on the Controller.                  |

## Documentation

### Start here

| You want to...                                           | Read...                                            |
|----------------------------------------------------------|----------------------------------------------------|
| Install, pair, and play your first game                  | <a href="#">Installation Guide</a>                 |
| Learn gestures, Programs, and game controls              | <a href="#">Gameplay Guide</a>                     |
| Understand joystick, merged-controller, and native modes | <a href="#">VirtualGlove Input Modes</a>           |
| Choose or troubleshoot a camera                          | <a href="#">Camera Guide</a>                       |
| Assemble a printed case quickly                          | <a href="#">Enclosure Assembly Quick Reference</a> |
| Print or customize a case                                | <a href="#">Controller Enclosure Guide</a>         |
| Recognize matrix animations and messages                 | <a href="#">Matrix Display Guide</a>               |
| Find a quick command or status reminder                  | <a href="#">Quick Reference</a>                    |
| Solve a problem by symptom                               | <a href="#">Troubleshooting</a>                    |

## Go deeper

| You want to...                                                       | Read...                                          |
|----------------------------------------------------------------------|--------------------------------------------------|
| See the current components and data flow                             | <a href="#">Architecture</a>                     |
| Look up every setting and command                                    | <a href="#">Configuration Reference</a>          |
| Follow the engineering process, experiments, and movement validation | <a href="#">Engineering Journey</a>              |
| Repeat camera, latency, trace, or native research                    | <a href="#">Engineering Toolkit</a>              |
| Review ROM-level native and joystick evidence                        | <a href="#">Power Glove Game ROM Input Audit</a> |
| Review security and pairing boundaries                               | <a href="#">Security Policy</a>                  |
| Check dependencies and third-party terms                             | <a href="#">Third-party Notices</a>              |
| Contribute code or documentation                                     | <a href="#">Contributing Guide</a>               |

Printable editions of all maintained guides are available in [output/pdf/](#). The Controller serves the same documentation from its local Help page.

## Project status

VirtualGlove 0.5.0 keeps the proven CPU MediaPipe Hands path, Programs 1-14, guided readiness checks, live dead-zone visualization, and source-accurate rapid-fire behavior. It extends the authenticated console integration to Recalbox, Batocera, and LaunchBox while retaining RetroPie.

The [v0.5.0-rc.1](#) candidate has completed physical controller and VirtualGlove acceptance on RetroPie, Recalbox 10.1.1, Batocera 43.1, and LaunchBox. This includes ordinary NES play, native Super Glove Ball, physical-controller coexistence, hotkeys, reboot persistence, and Controller Router remapping.

Recalbox and Batocera use Controller Router to publish enabled merged Players 1-4 from configured physical sources and at most one VirtualGlove. Standard RetroPie keeps its separate gamepad until Router is explicitly enabled; the project arcade cabinet now uses Router for its two I-PAC interfaces, 8BitDo controllers, and VirtualGlove. LaunchBox uses a local RetroPad while leaving XInput and real keyboard controls available. Every platform supports registered FCEUmm games and the separately named Nestopia (VirtualGlove) path for native Super Glove Ball. The installers update managed files while preserving pairing, players, calibration, tuning, Academy progress, games, ROMs, and saves.

Use Setup's **Download system report** when asking for help. It records useful software, camera, controller, and connection health without including video, pairing keys, player calibration, ROM names, or network addresses.

## Contributing and licensing

Issues, careful test reports, documentation improvements, and code contributions are welcome. Please read the [Contributing Guide](#) before opening a change. Release history is kept in the [Changelog](#).

VirtualGlove is maintained by **Iain Bennett** and is licensed under the [MIT License](#). The modified Nestopia core is GPLv2 software and remains separate from the MIT application. Nintendo, NES, Power Glove, and the named games belong to their respective owners. See [Third-party Notices](#) for dependency, model, asset, and native-core licensing details.